

Project Launch

Procurement Spend Analysis and Supplier Optimization for FMCG operations in Nigeria.

- Analyzed ~~₦~~310.4B procurement spend across 24 months, 2,500 orders, and 40 suppliers.
- Built an end-to-end analytics pipeline: data generation, SQL model, optimization, and dashboard.
- Objective: convert transaction data into ranked, quantified decisions.

Live dashboard: procurementspendanalysis.streamlit.app

What Problem Does It Solve?

Most teams can report transactions, but cannot explain where money leaks or what to fix first.

- Spend visibility gaps across categories and suppliers.
- Inconsistent pricing for the same material across teams.
- Late-delivery and quality failures hidden from invoice-level reporting.
- Maverick buying and FX exposure not quantified for leadership decisions.

From record-keeping to decision intelligence.

How It Works

Five-layer system from raw files to executive-ready insights.

- 1) Generate realistic procurement data (suppliers, materials, POs, incidents).
- 2) Load into SQLite and build analytics views.
- 3) Compute KPIs and savings opportunities in Python + SQL.
- 4) Run optimization, scenarios, and Monte Carlo uncertainty.
- 5) Publish outputs in a 5-page Streamlit dashboard.

Tech stack: Python, SQL, SQLite, Streamlit, Plotly.

Key Findings

The numbers revealed high-value actions already available in existing data.

- Total identified savings: ~~¥~~185.9B (59.9% of spend).
- Price standardization opportunity: ~~¥~~18.5B.
- Supplier performance opportunity: ~~¥~~167.5B.
- Maverick spend exposure: ~~¥~~40.6B (13.08%).
- USD exposure: \$132.4M with 99.84% FX volatility.

Insight: performance losses outweighed pure price inefficiencies.

Decision Logic

Supplier decisions are ranked with a transparent weighted model.

- Composite score weights: cost 45%, delivery 30%, quality 15%, risk 10%.
- Top suppliers are selected per category and assigned recommended share.
- Constraints enforce minimum performance and risk thresholds.
- Result: lower total cost, better resilience, and clearer governance.

Optimization turns strategy into executable sourcing actions.

Planning Under Uncertainty

Single-point savings claims are risky. This project includes confidence ranges.

- Scenario analysis: Conservative, Base, and Aggressive outcomes.
- Monte Carlo simulation: 10,000 runs for uncertainty quantification.
- Outputs include P05, median, and P95 savings bounds for realistic planning.
- Supports CFO-grade budgeting and risk-adjusted target setting.

Use median for planning and percentile bounds for contingency.

Decisions You Can Make

Built to drive action across Finance, Procurement, and Operations.

- Prioritize renegotiation by highest-value categories.
- Replace or remediate underperforming suppliers.
- Cut maverick spend through stronger vendor governance.
- Set FX hedging with quantified USD exposure and volatility.

Every recommendation is tied to expected financial impact.

Launch and Access

Procurement Spend Analysis is now live and open for review.

- Live app: procurementspendanalysis.streamlit.app
- Code: github.com/DavidMaco/Procurement_Spend_Analysis
- Includes dashboard, reports, optimization outputs, and test suite.
- Open to feedback from procurement, analytics, and finance leaders.